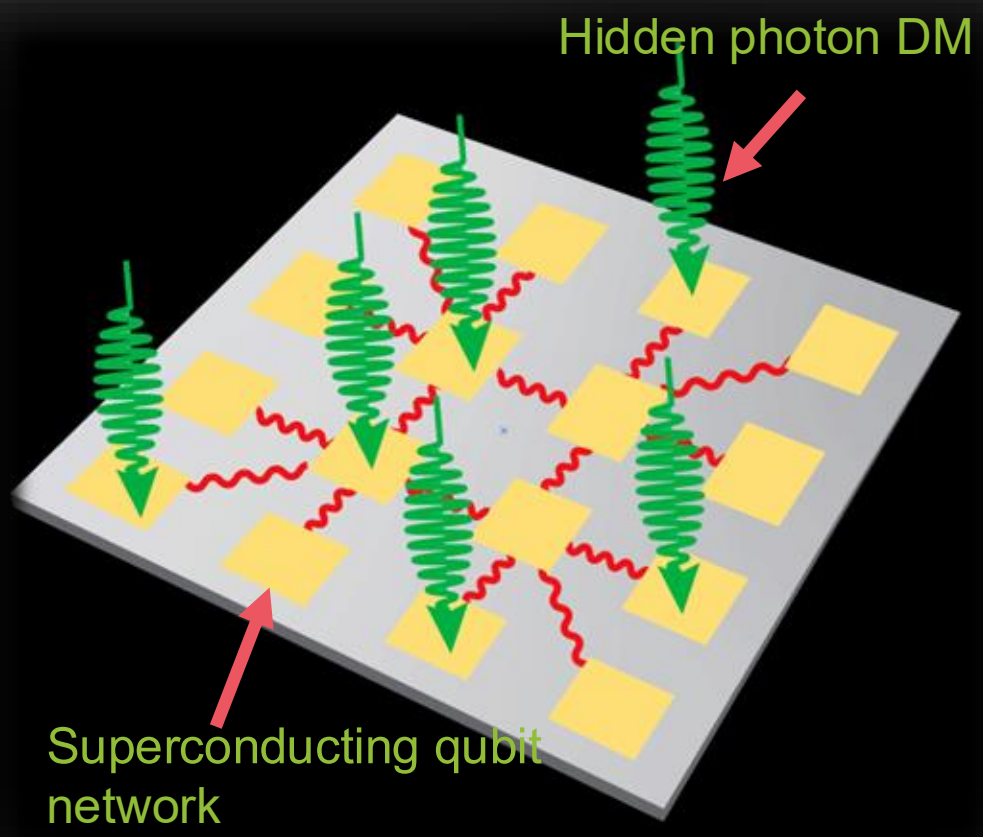
Dark matter detection with superconducting sensor network

Le Bin Ho

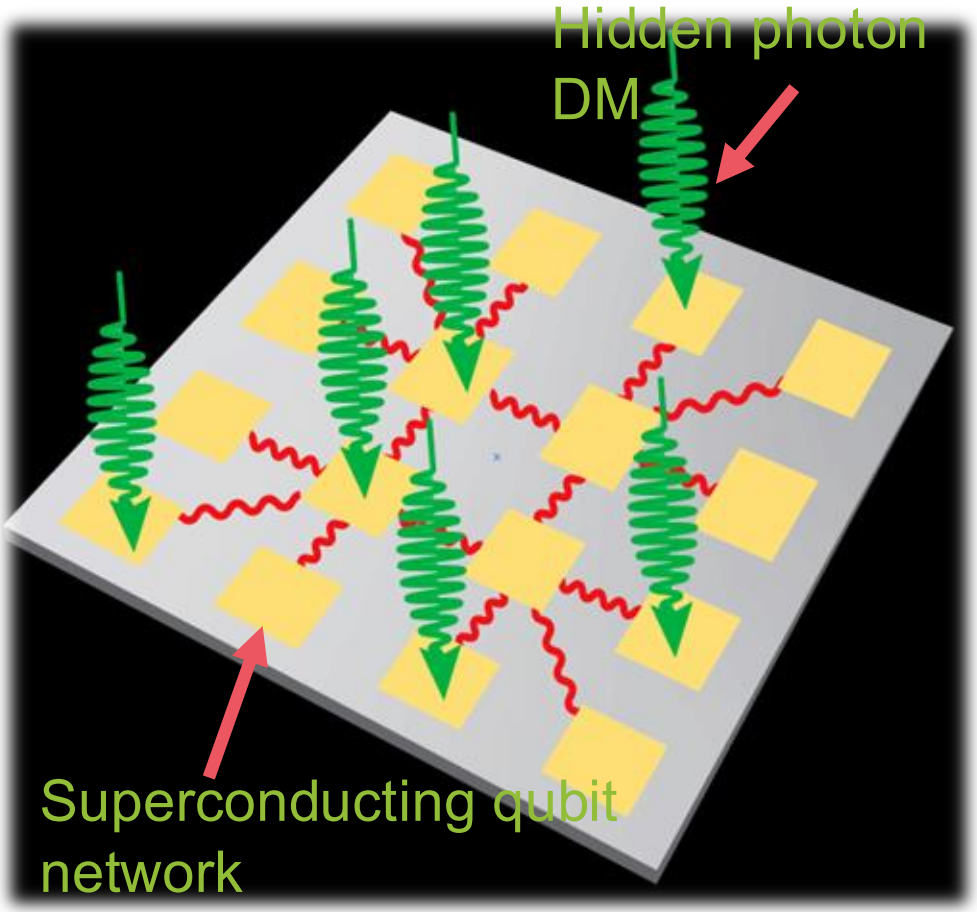
Tohoku University, Japan



Adriel I Santoso, Le Bin Ho, Physical Review D 112, L081301 (2025)

Outline

- **Dark matter search overview**
- **Qubit-based quantum sensors**
- **Quantum sensor network for DM**



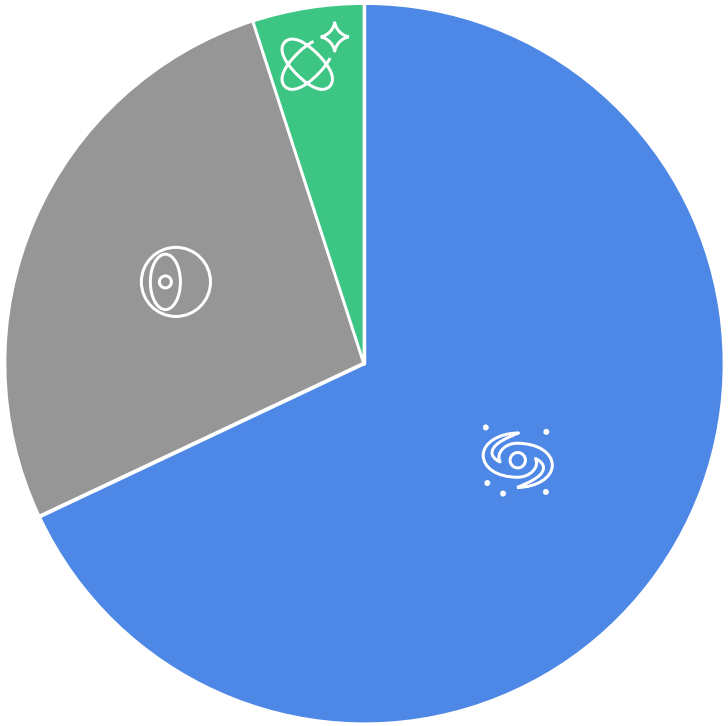
Composition of the Universe

5% Ordinary Matter

Atoms and particles that make up stars, planets, gas, dust, and us.

27% Dark Matter

Invisible matter detected by its gravitational effects.



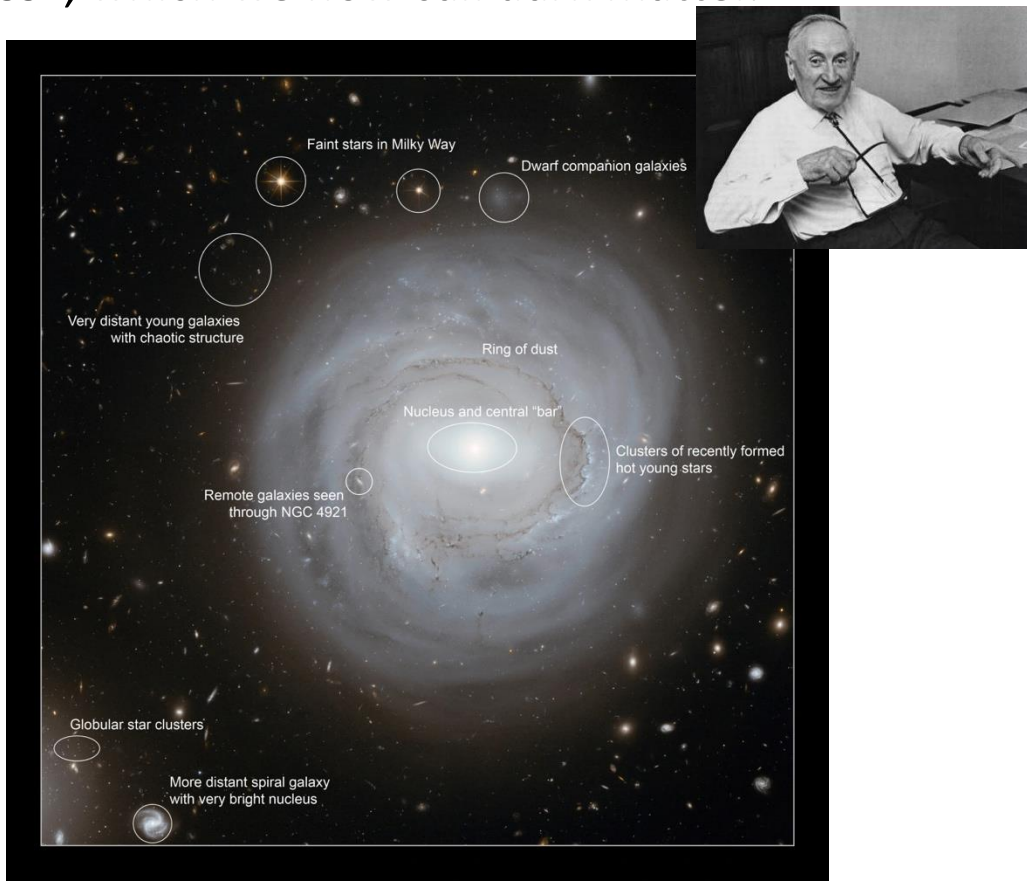
68% Dark Energy

A mysterious force causing the accelerated expansion of the universe.

Very first hints of dark matter

1930s – Fritz Zwicky

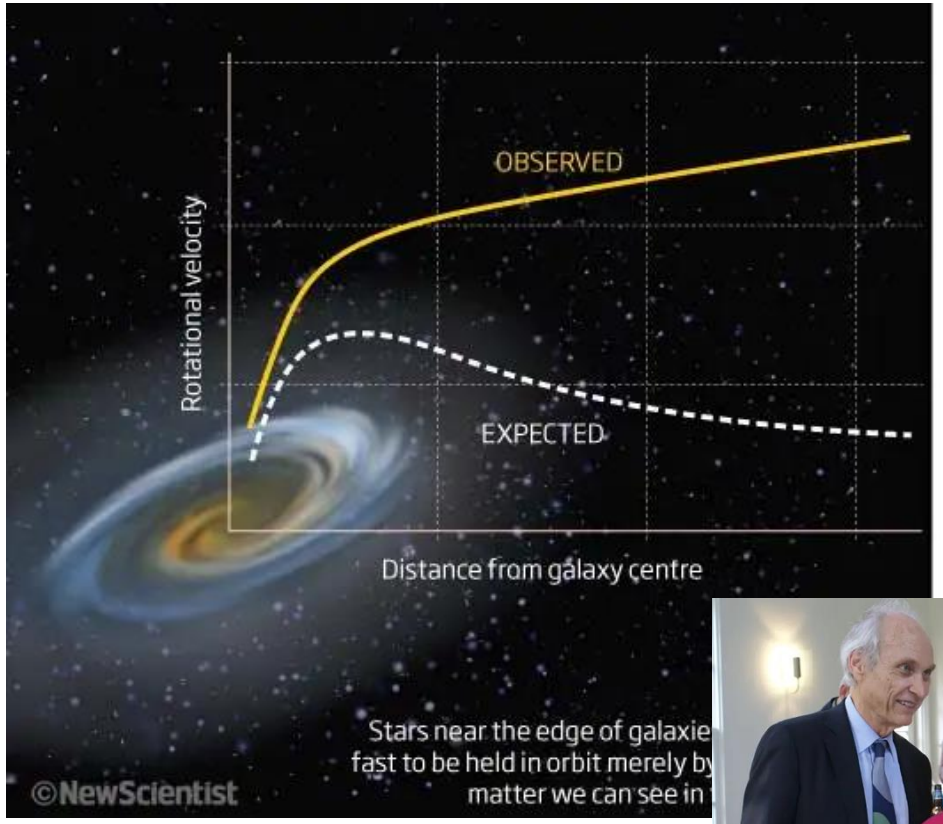
He studied the motion of galaxies in the Coma Cluster and found they moved too fast to be held together by visible matter. He proposed the existence of **"missing mass"**, which we now call dark matter.



Coma Cluster / Credit: NASA

1970s – Vera Rubin and Kent Ford

They observed that stars in spiral galaxies orbit at roughly the same speed regardless of distance from the center—a behavior that suggests the presence of an **invisible mass halo**.



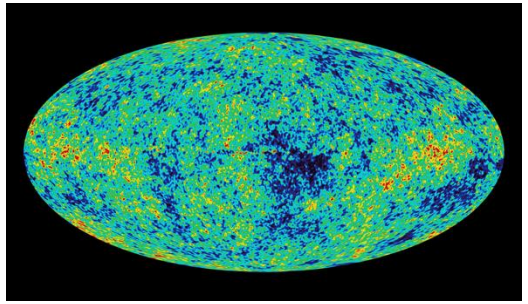
Credit: NewScientist

Credit: APS

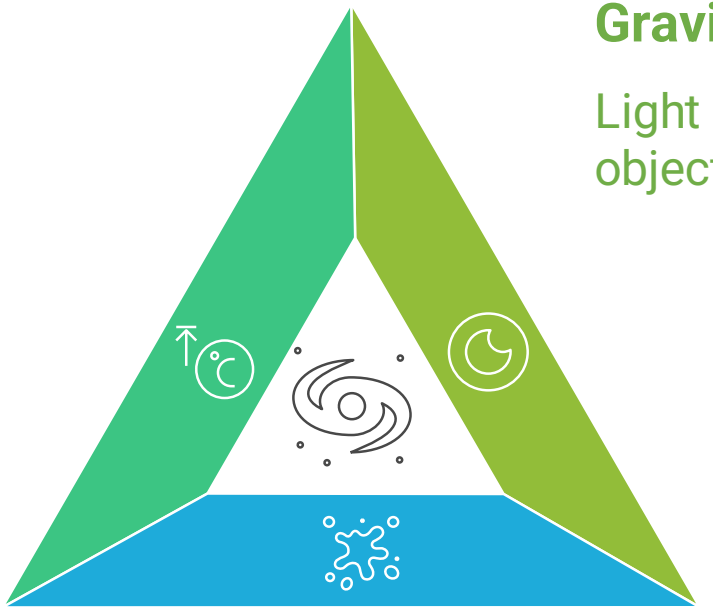
Modern Era

Cosmic Microwave Background (CMB)

Temperature fluctuations in the early universe, indicating dark matter's presence

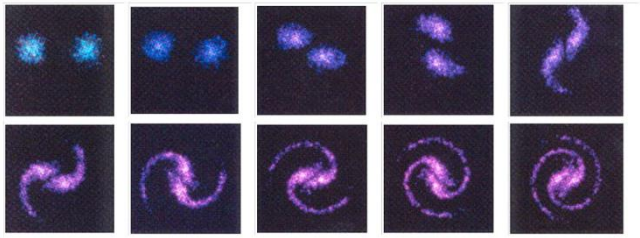


WMAP image of the CMB
Credit: wiki



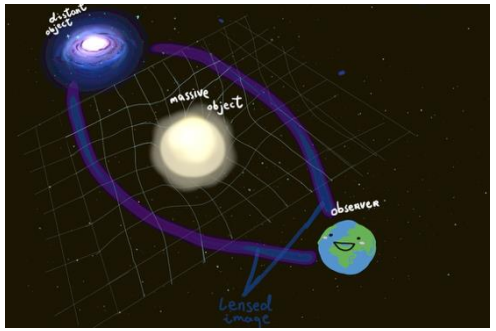
Galaxy Formation Models

Simulations showing dark matter's role in galaxy structure



Gravitational Lensing

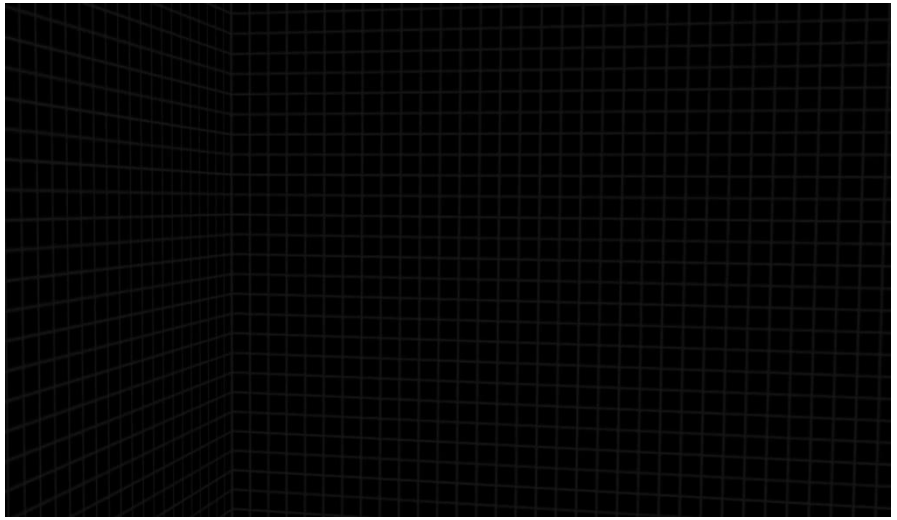
Light bending around massive objects, revealing hidden mass



Seeing the universe through dark matter lenses
Credit: palatinate.org.uk

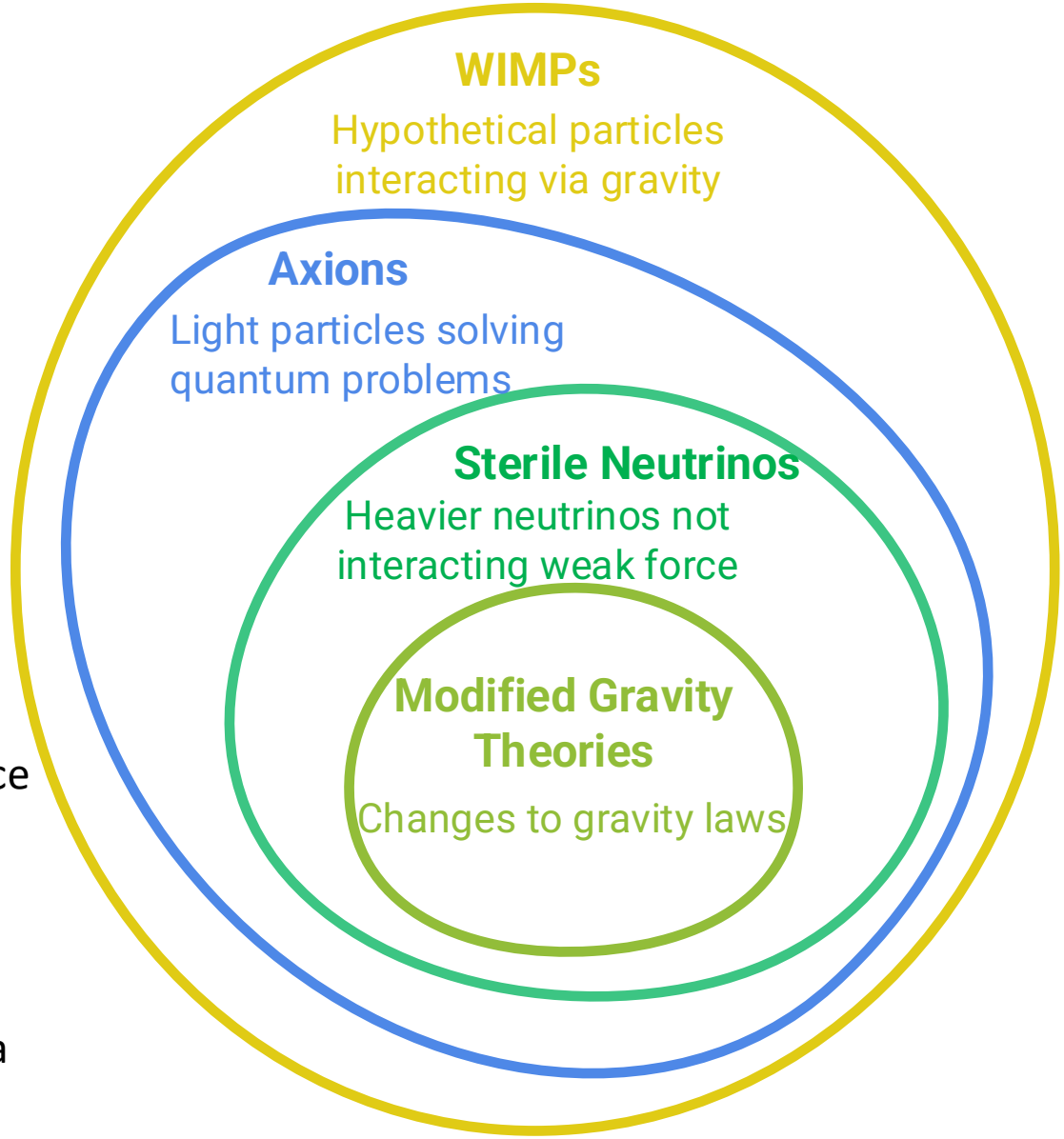
Leading Theories

Weakly interacting massive particles (WIMPs)



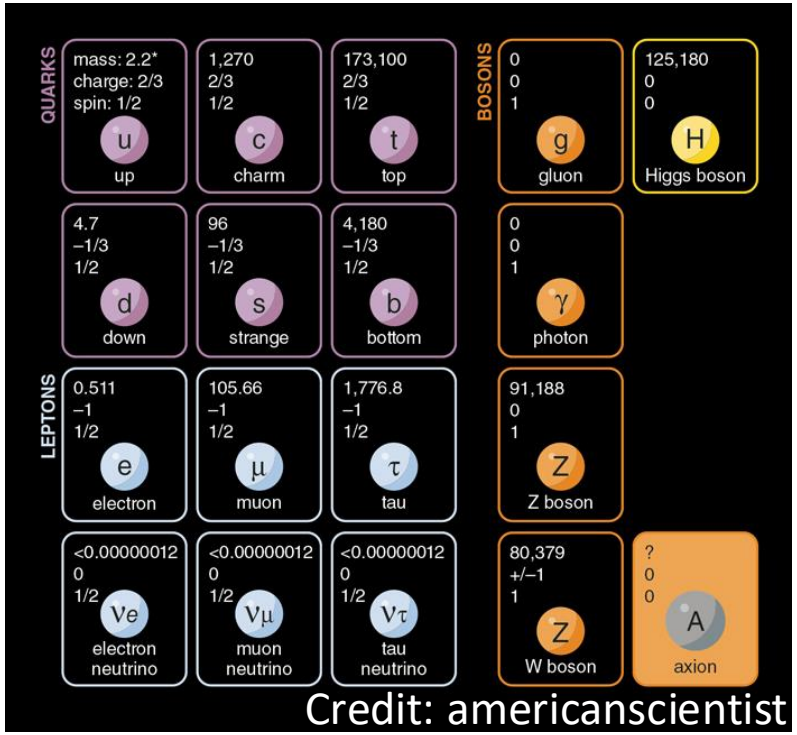
Credit: NASA

- Hypothetical particles interacting via gravity and weak force
- No electromagnetic interaction → invisible to light-based detection
- **Indirect detection**: search for annihilation/decay products (e.g., gamma rays, positrons)
- **Direct detection**: rare WIMP collisions in large detectors (a few events/year expected)



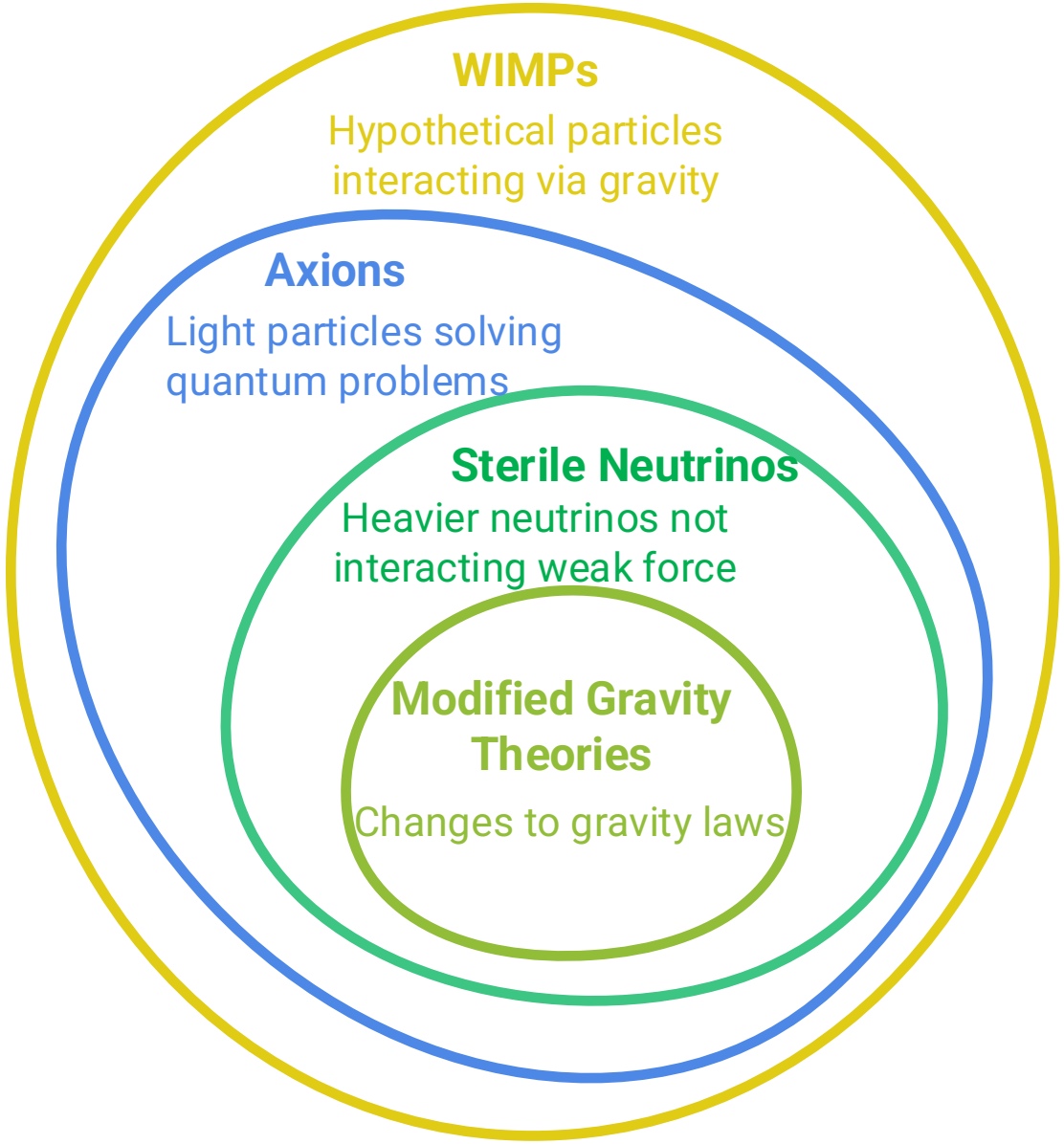
Leading Theories

Axions

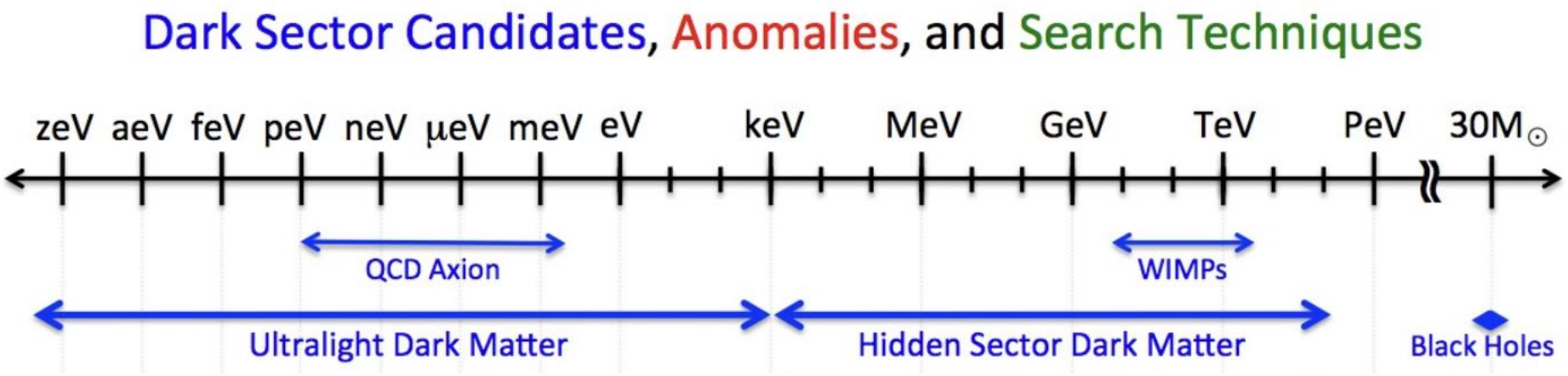


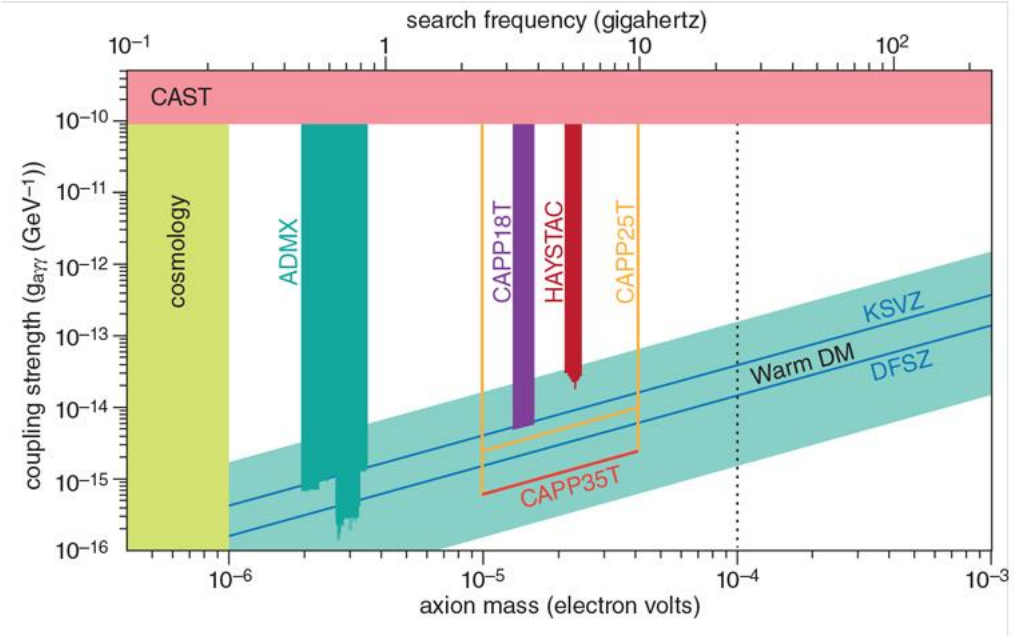
Credit: americanscientist

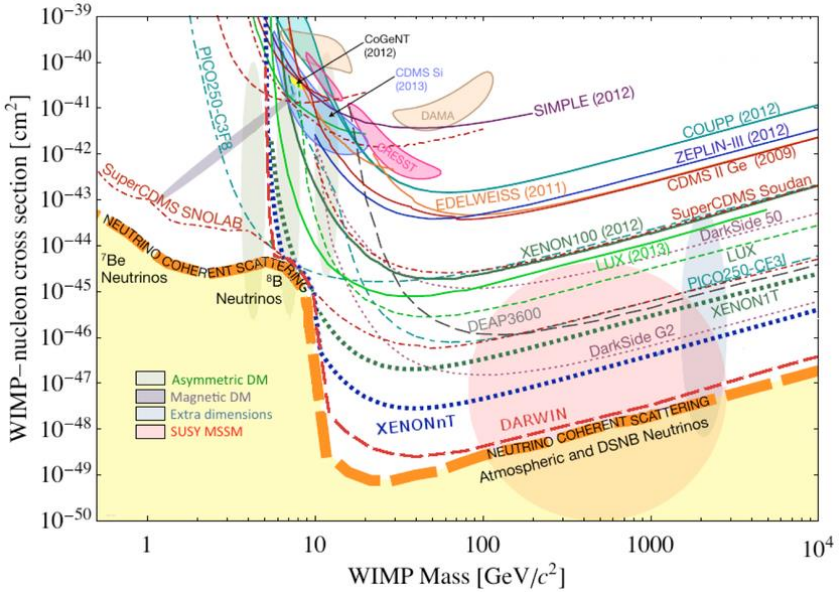
- Hypothetical particles solving the strong CP problem in QCD.
- Ultra-light, stable, weakly interacting → dark matter candidate.
- Detection via axion–photon conversion in strong magnetic fields.
- Key experiments: **ADMX**, **CASPEr**, **MADMAX**



Leading Theories



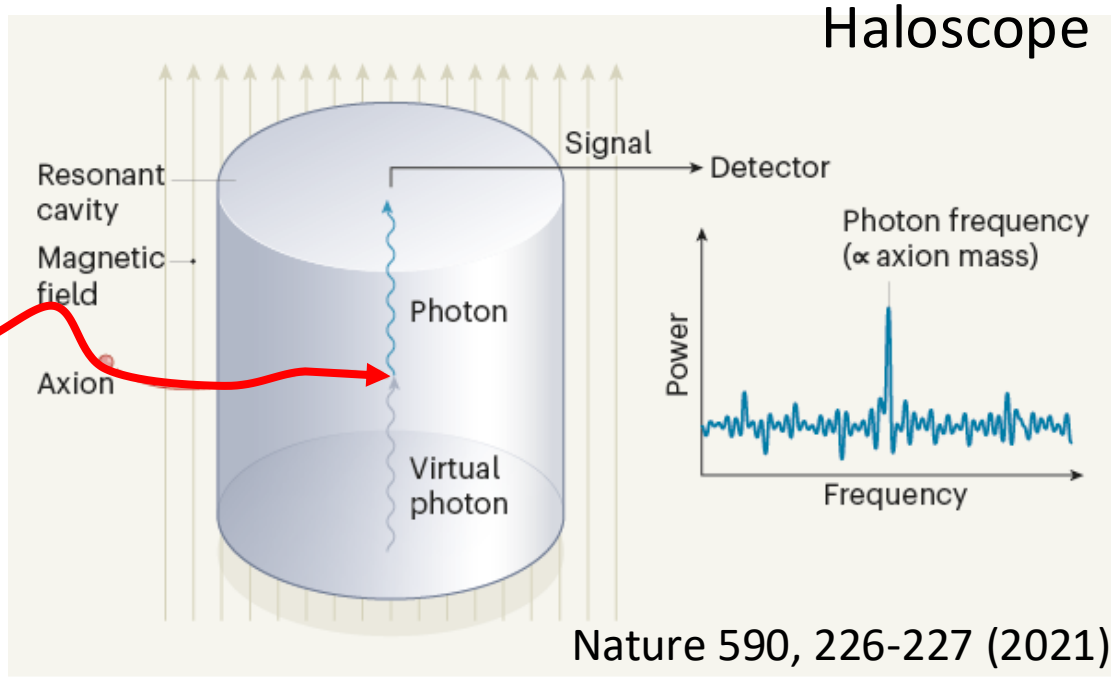




PhysRevLett.133.211803 (2024)

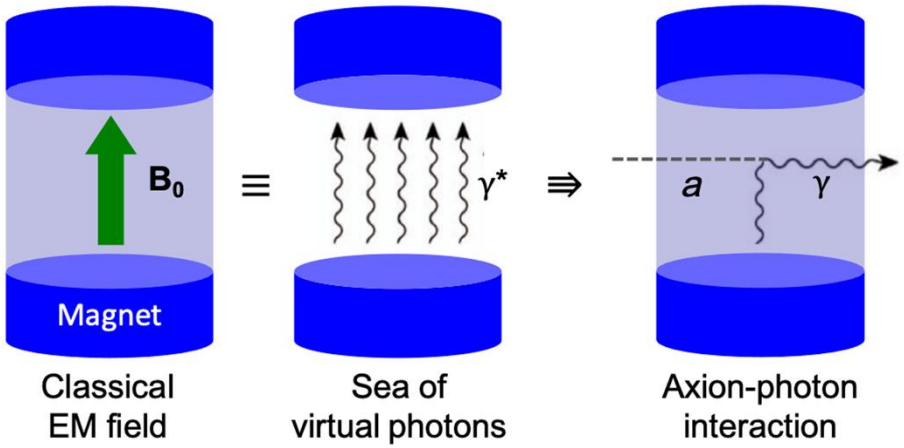
Axion–Photon Interaction and DM detection

Credit: google

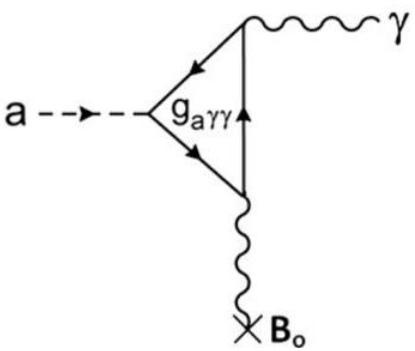


Nature 590, 226-227 (2021)

Sci. Adv. 8, eabm9928 (2022)



$$\mathcal{L}_{a\gamma\gamma} = -\frac{1}{4} g_{a\gamma\gamma}\, a\, F_{\mu\nu}\tilde{F}^{\mu\nu} = g_{a\gamma\gamma}\, a\, \mathbf{E}\cdot\mathbf{B}$$



- $g_{a\gamma\gamma}$ is the axion-photon coupling constant,
- $F_{\mu\nu}$ is the electromagnetic field tensor,
- $\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\alpha\beta}F_{\alpha\beta}$ is the dual tensor,
- $\mathbf{E}$ and $\mathbf{B}$ are the electric and magnetic fields.

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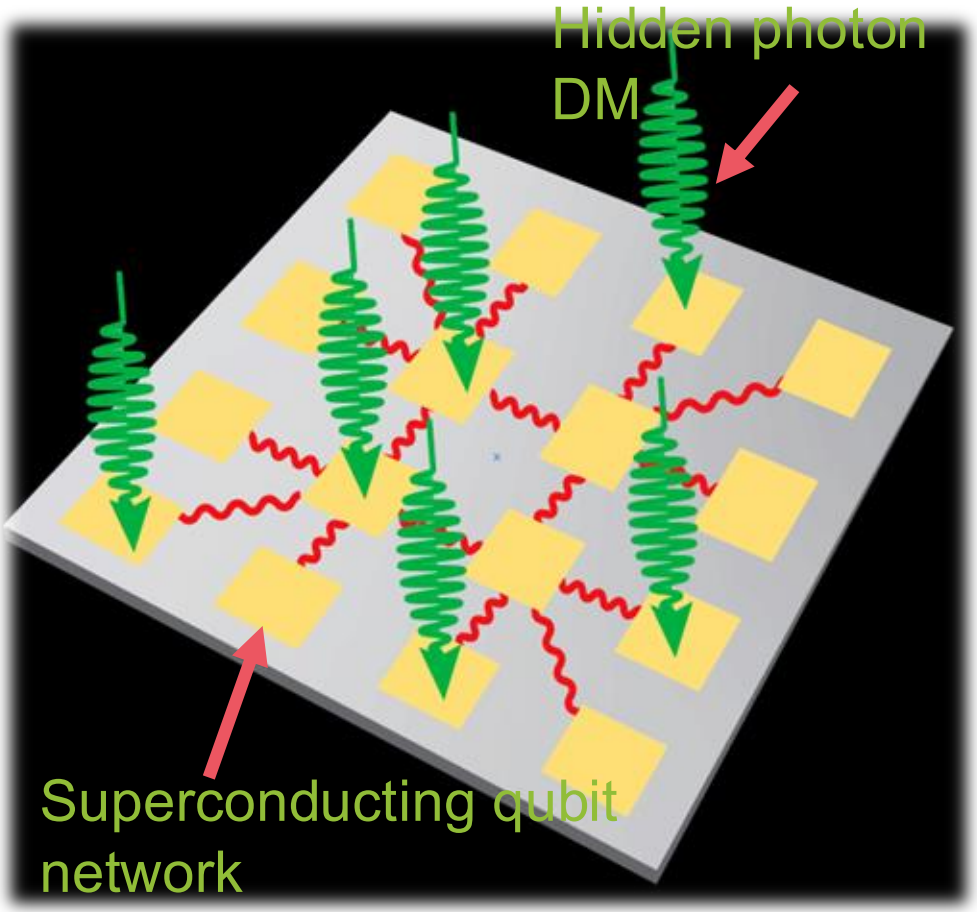
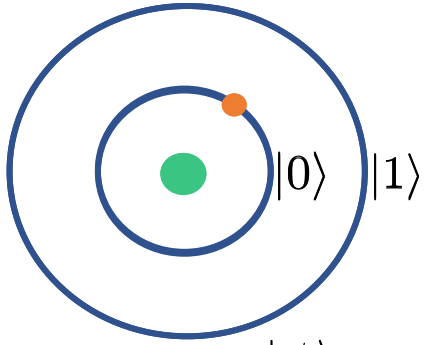


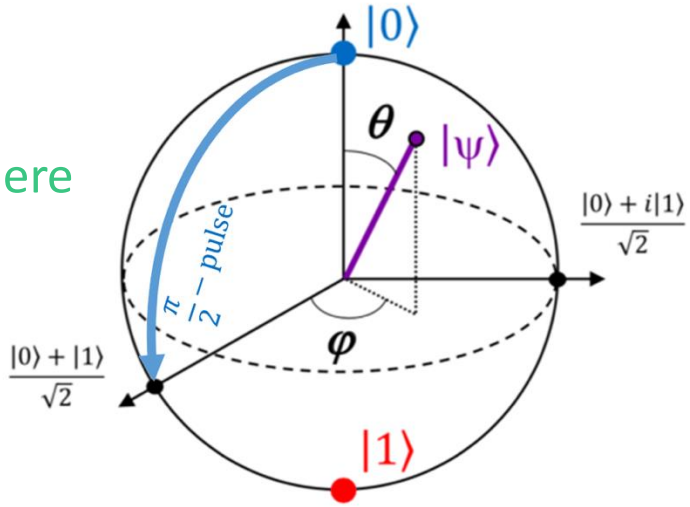
Superconducting qubit



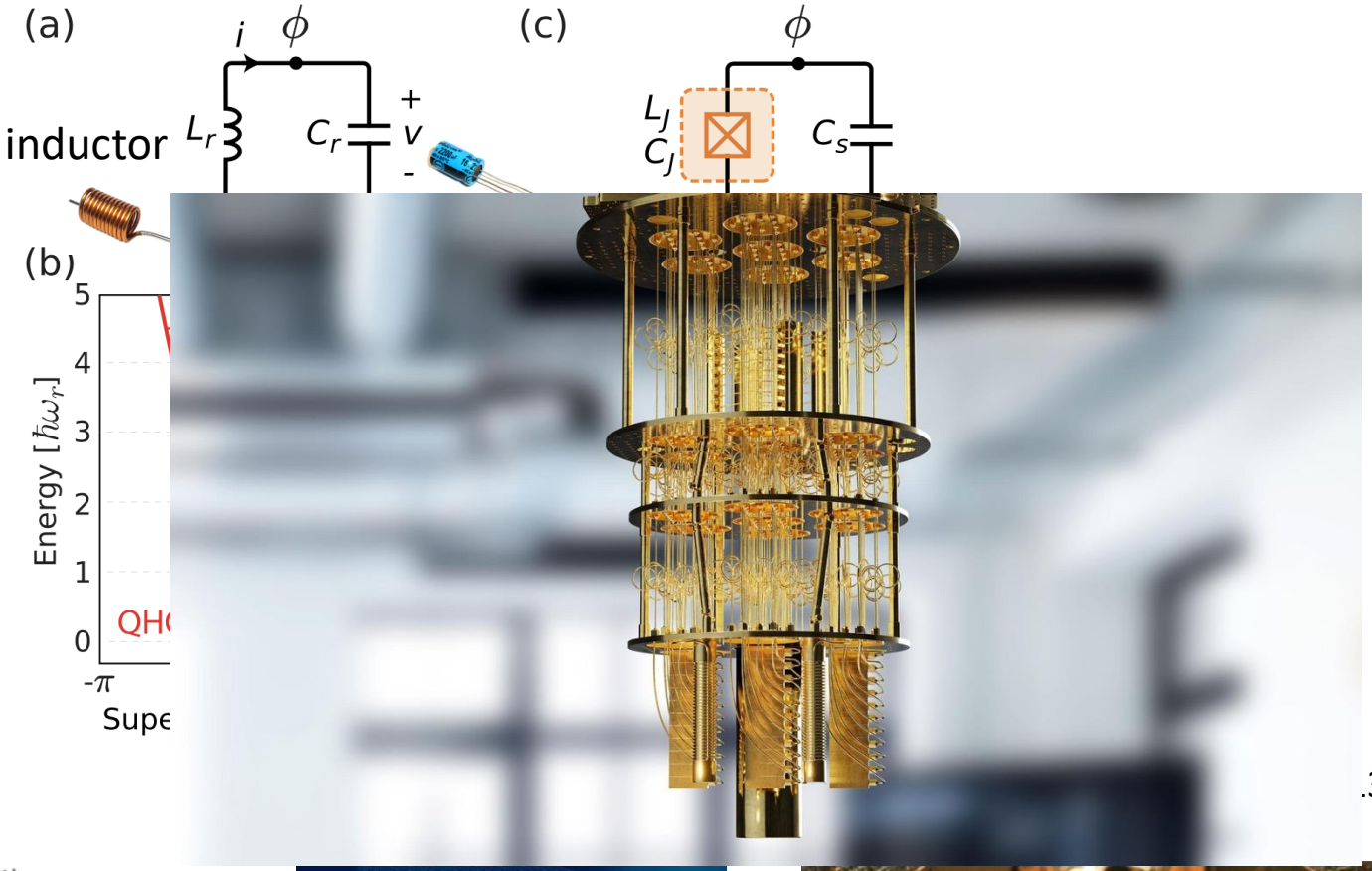
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

- a two-level system
- normalize, $|\alpha|^2 + |\beta|^2 = 1$

Bloch sphere



$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle$$



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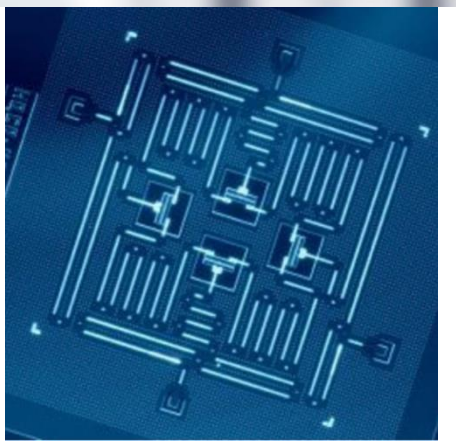
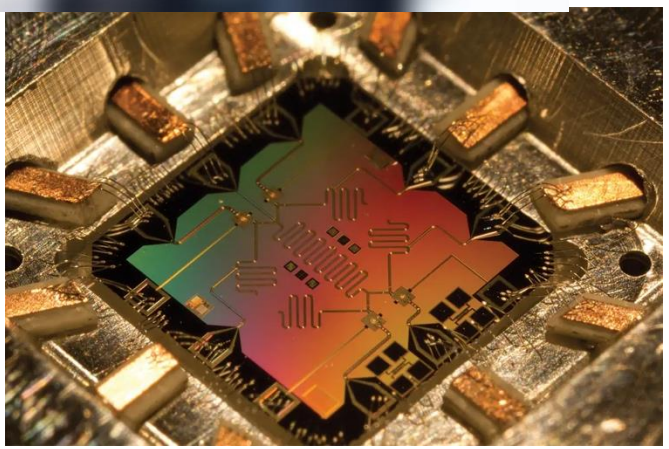


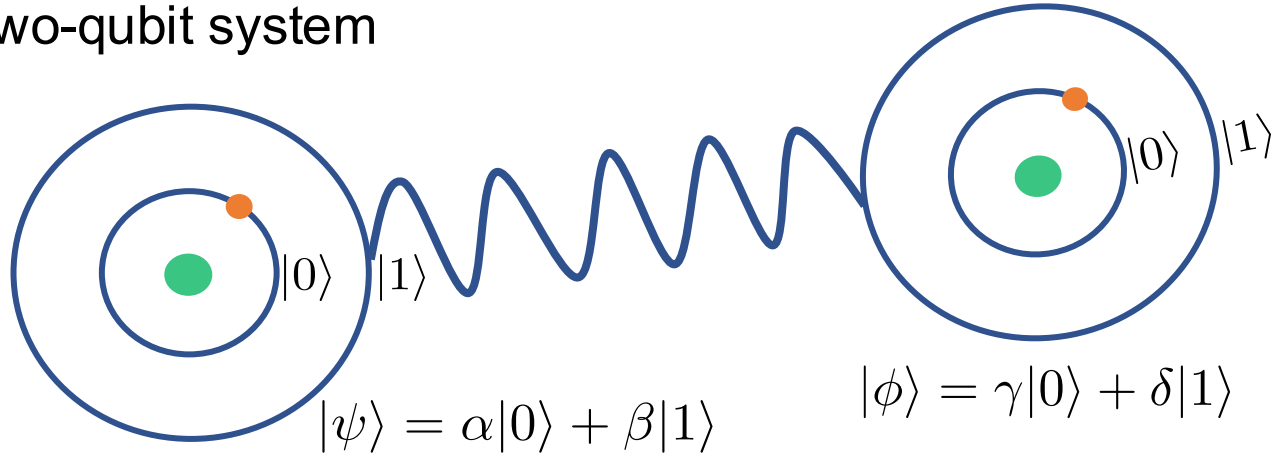
Transmon qubit/wikipedia



Google

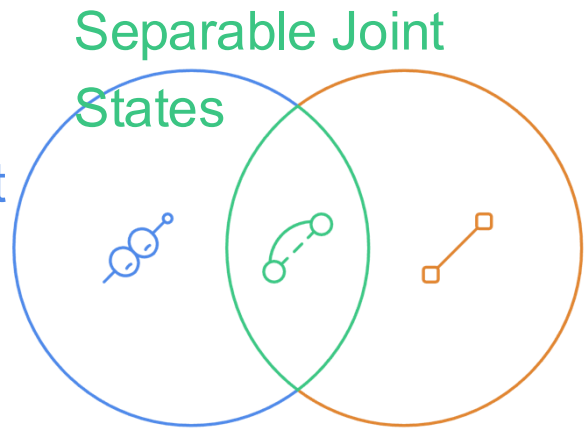
Entanglement

Two-qubit system



$$|\Psi\rangle = |\psi\rangle \otimes |\phi\rangle = \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle$$

Entangle: cannot separate the joint state into tensor product.

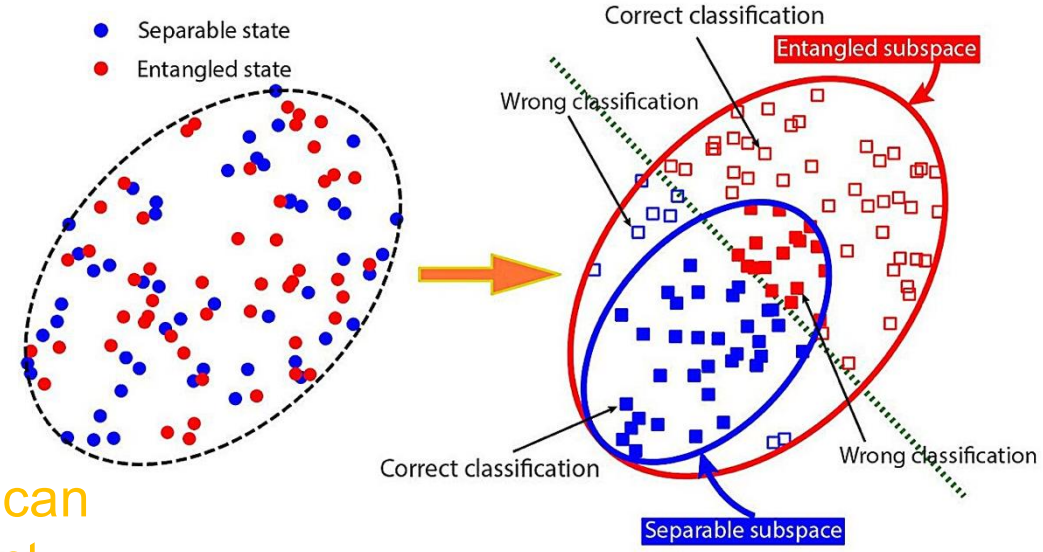


Non-entangle: can separate the joint state into tensor product.

e.g., a GHZ state $|\Psi\rangle = \frac{1}{\sqrt{2}}\Big(|00\cdots0\rangle + |11\cdots1\rangle\Big)$

Variational entanglement witness (VEW)

PhysRevResearch.7.013239(2025



$$\mathcal{C}(\boldsymbol{\alpha}) = \mathrm{Tr}[\mathcal{W}(\boldsymbol{\alpha})\rho].$$

The optimization process is defined by

$$\boldsymbol{\alpha}^* = \arg\min_{\{\boldsymbol{\alpha}\}} \mathcal{C}(\boldsymbol{\alpha})$$

$$\text{s.t.} \quad \mathcal{C}(\boldsymbol{\alpha}^*) = 0 \quad \forall \text{ separable states } \rho_{\text{sep}}.$$

Axion–Qubit Interaction and DM detection

$$\mathcal{H} = \omega|e\rangle\langle e| - 2\eta\cos\left(m_{\rm DM}t - \alpha\right)\left(|e\rangle\langle g| + |g\rangle\langle e|\right),$$

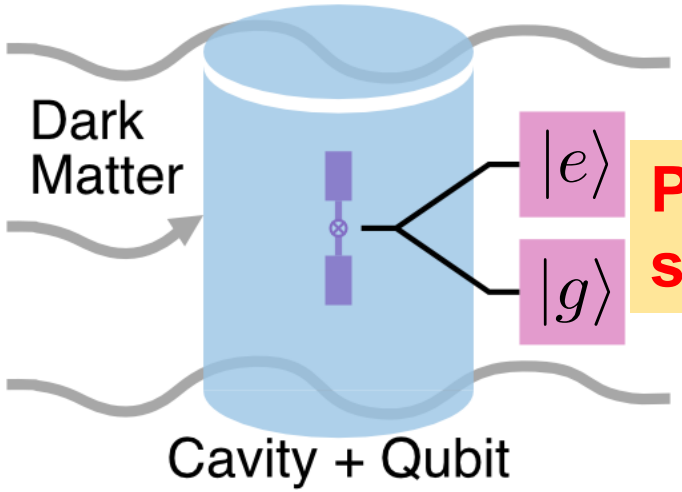
ω energy splitting

$m_{\rm DM}$ DM mass

α DM offset

coupling strength

$$\eta \equiv \frac{\sqrt{C\omega}}{2\sqrt{2}}\bar{E}^{(\rm eff)} d\cos\Theta,$$



Credit: FermiLab

Problem: single qubit does not give high sensitivity

$$\begin{pmatrix}\psi_g(\tau)\\ \psi_e(\tau)\end{pmatrix} = U_{\rm DM}(\tau)\begin{pmatrix}\psi_g(0)\\ \psi_e(0)\end{pmatrix}$$

$$U_{\rm DM}(\tau) = \begin{pmatrix}\cos\delta & ie^{-i\alpha}\sin\delta\\ ie^{i\alpha}\sin\delta & \cos\delta\end{pmatrix}, \quad \delta \equiv \eta\tau,$$

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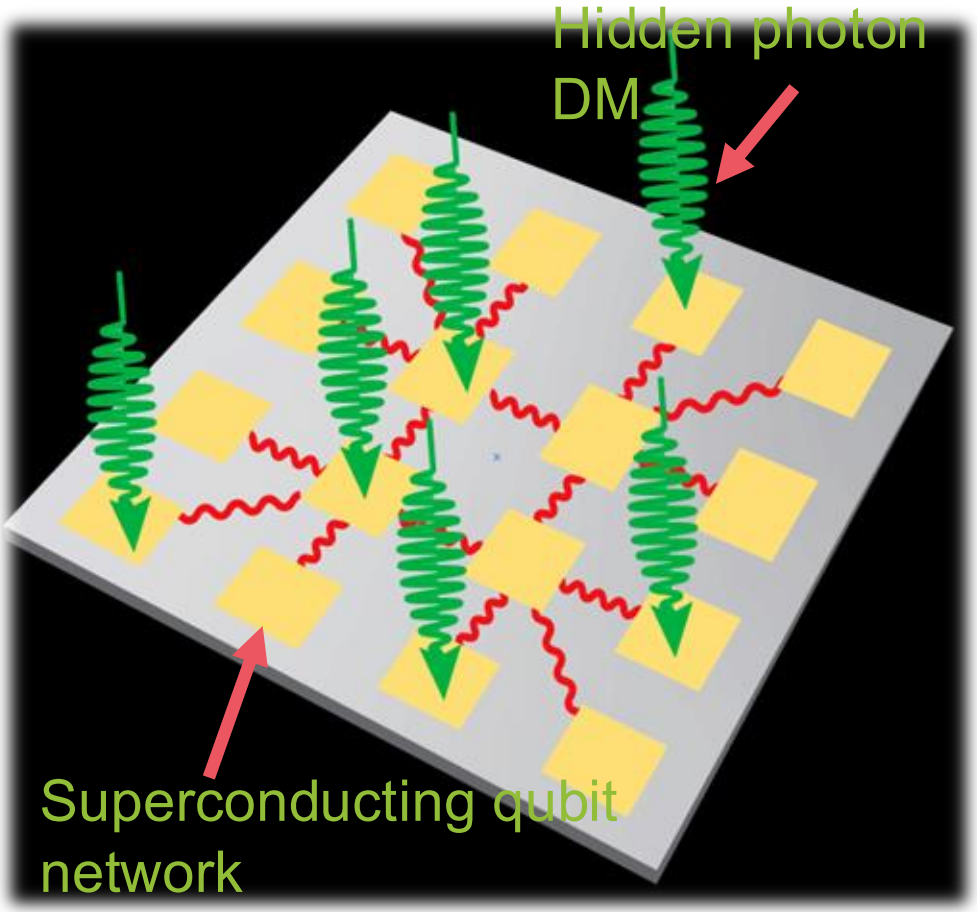
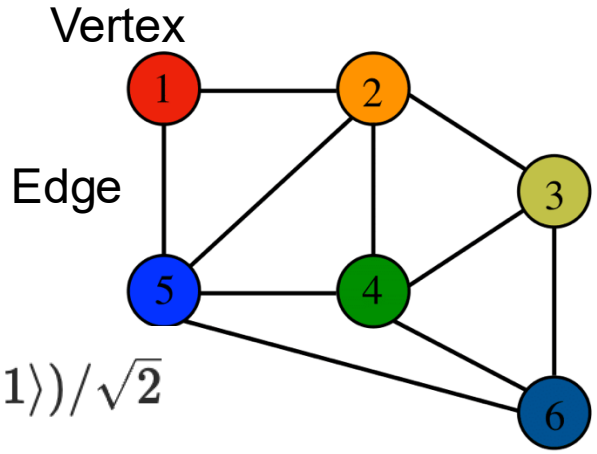


Sensor(qubit) network

Graph-like network $G = (V, E),$



$$|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

$$\mathrm{CZ} = \mathrm{diag}(1, 1, 1, -1).$$

$$|G\rangle = \left[\prod_{e\in E} \mathrm{CZ}_e\right] |+\rangle^{\otimes |V|}.$$

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

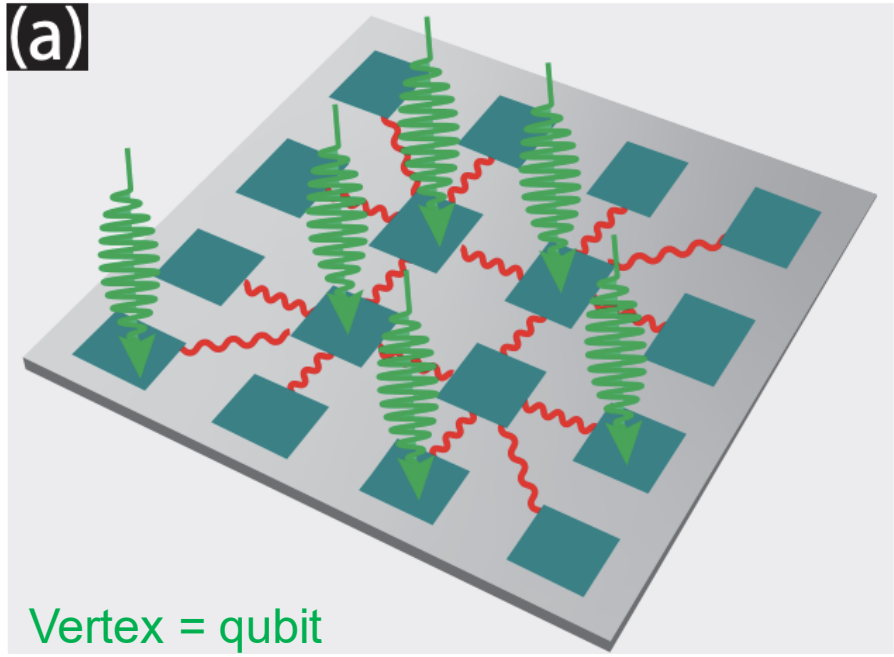
$$|+\rangle|+\rangle \longrightarrow CZ|+\rangle|+\rangle \longrightarrow HHCZ|+\rangle|+\rangle$$

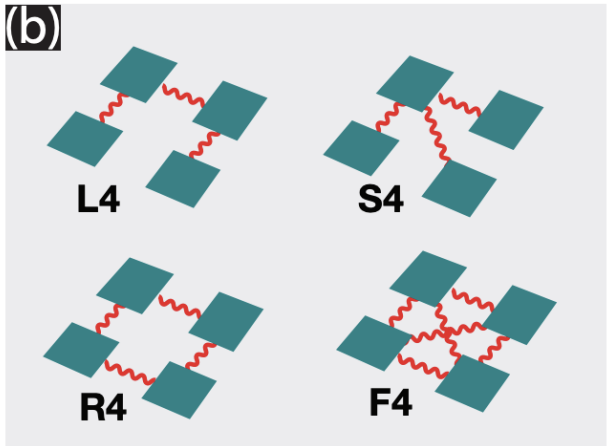
$$\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

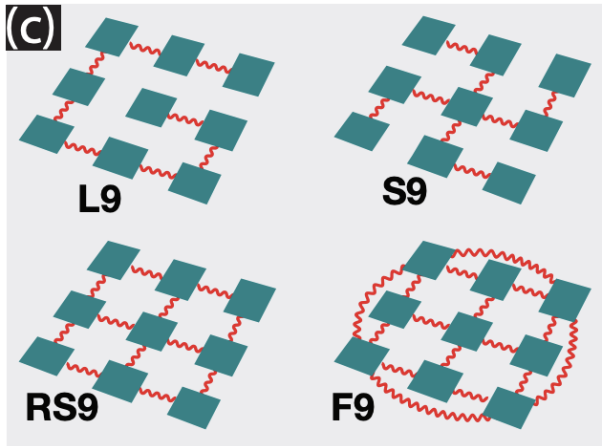
(separable state)

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

(entangled state)

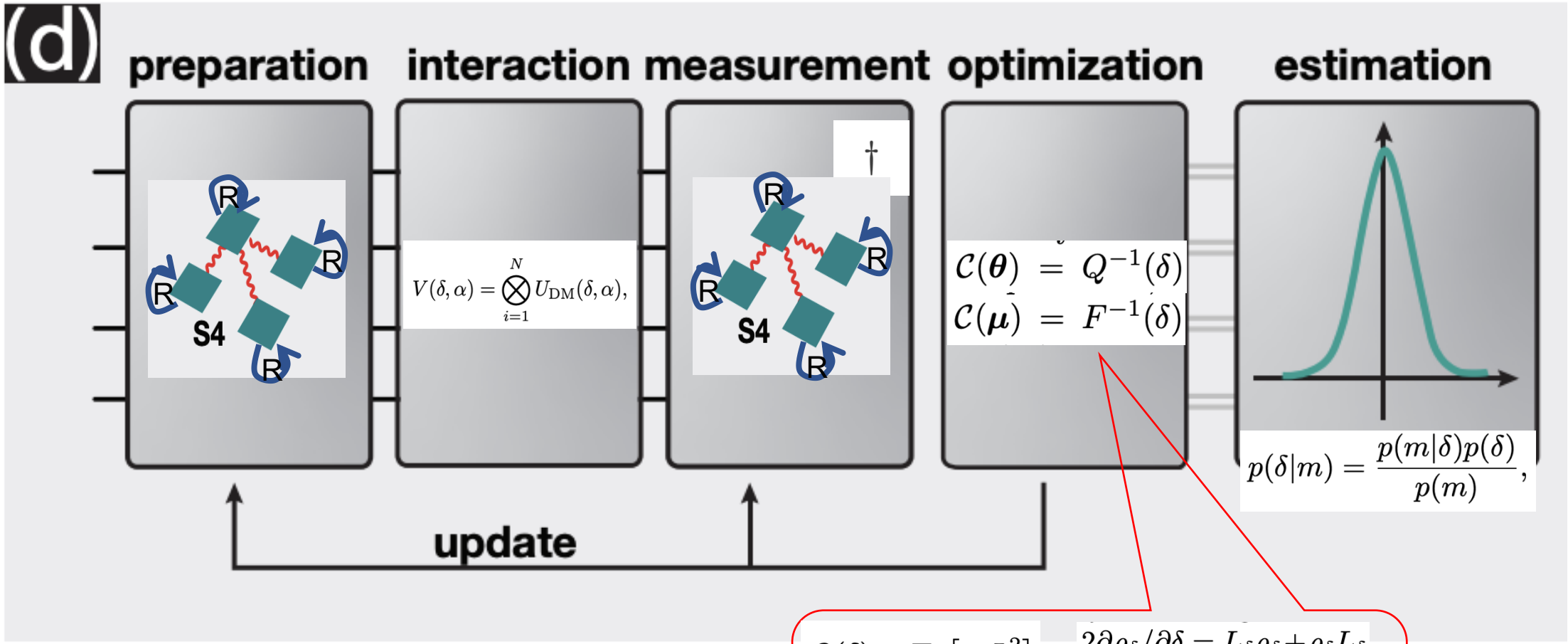






DM detection sensitivity in the sensor network

Optimization scheme



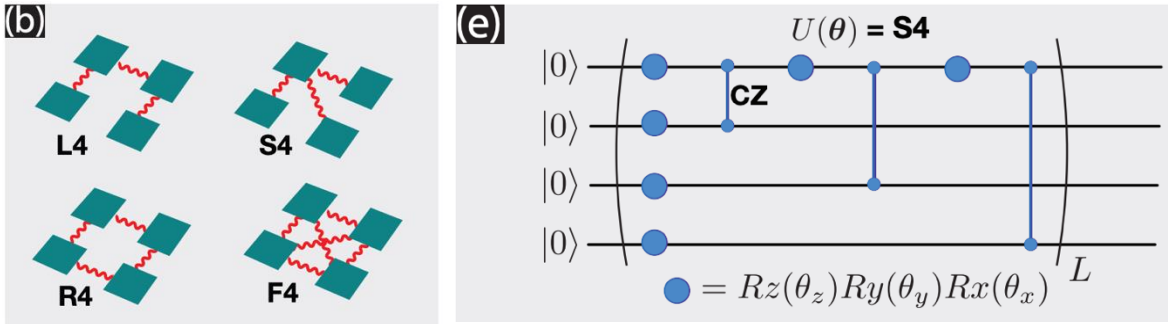
$$Q(\delta) = \mathrm{Tr}\left[\rho_\delta L_\delta^2\right] \qquad 2\partial\rho_\delta/\partial\delta = L_\delta\rho_\delta + \rho_\delta L_\delta$$

$$F(\delta) = \sum_m \frac{1}{p(m|\delta)}\left(\frac{\partial p(m|\delta)}{\partial\delta}\right)^2$$

Quantum bound

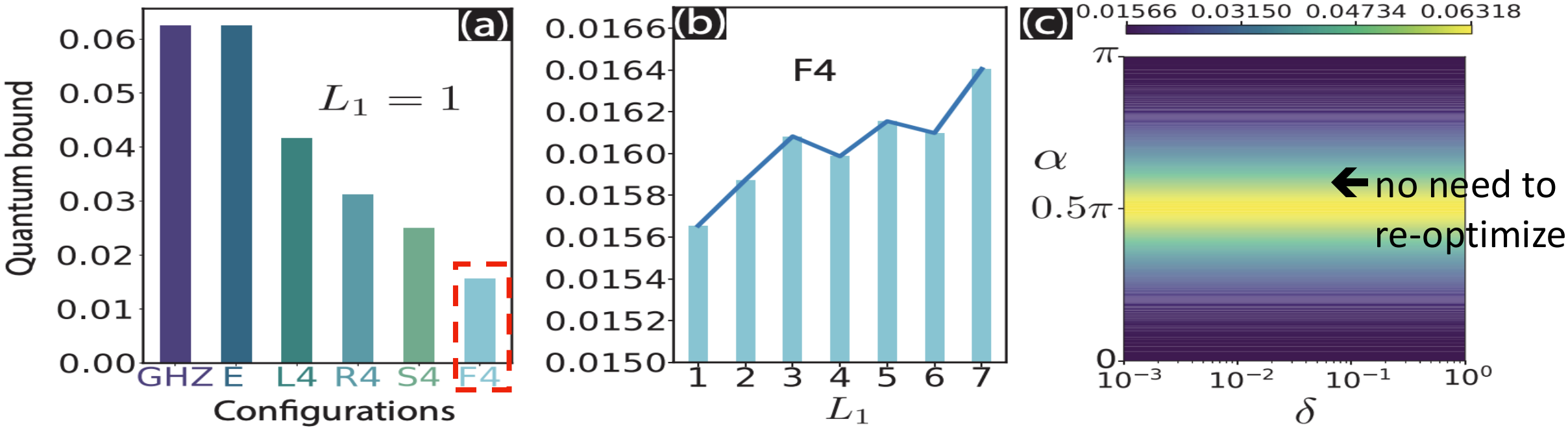
- Variance has the ultimate quantum bound

$$\nu\,\mathrm{Var}(\delta) \geq \frac{1}{F(\delta)} \geq \frac{1}{Q(\delta)},$$

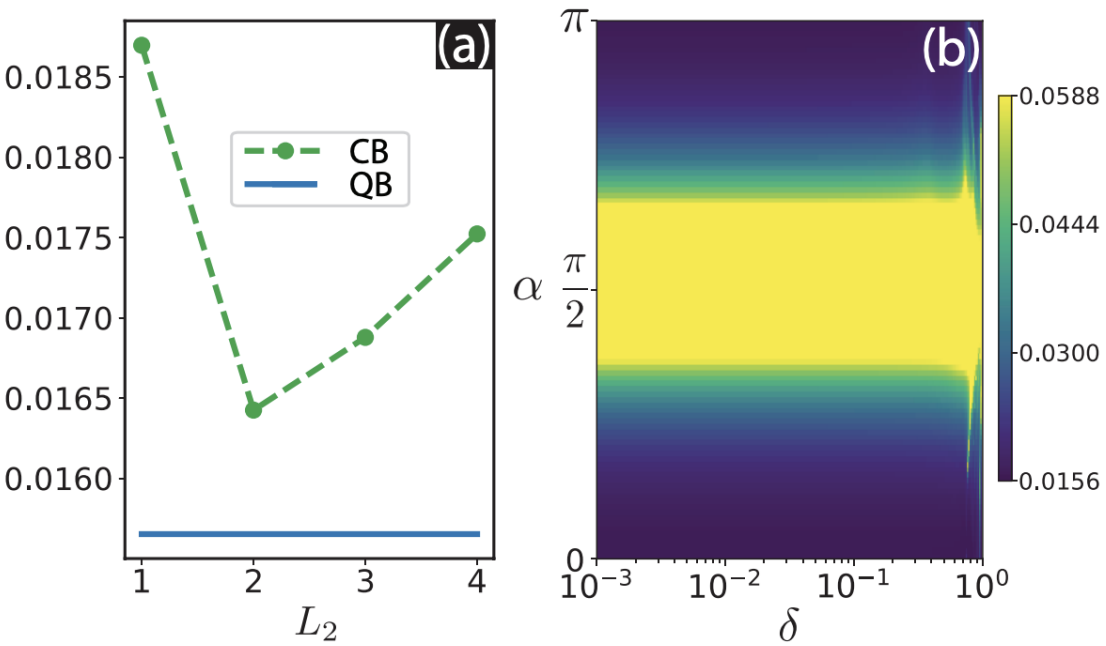


$$U_{\mathrm{DM}}(\tau) = \begin{pmatrix} \cos\delta & ie^{-i\alpha}\sin\delta \\ ie^{i\alpha}\sin\delta & \cos\delta \end{pmatrix}$$

$$\mathcal{C}(\boldsymbol{\theta}) = Q^{-1}(\delta)$$



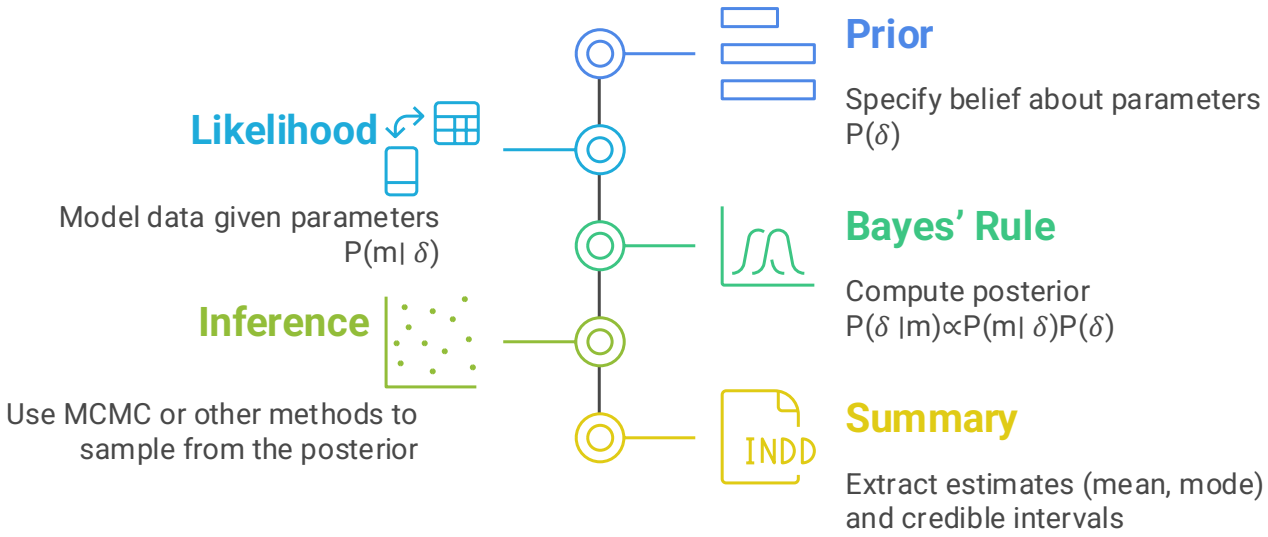
Classical bound

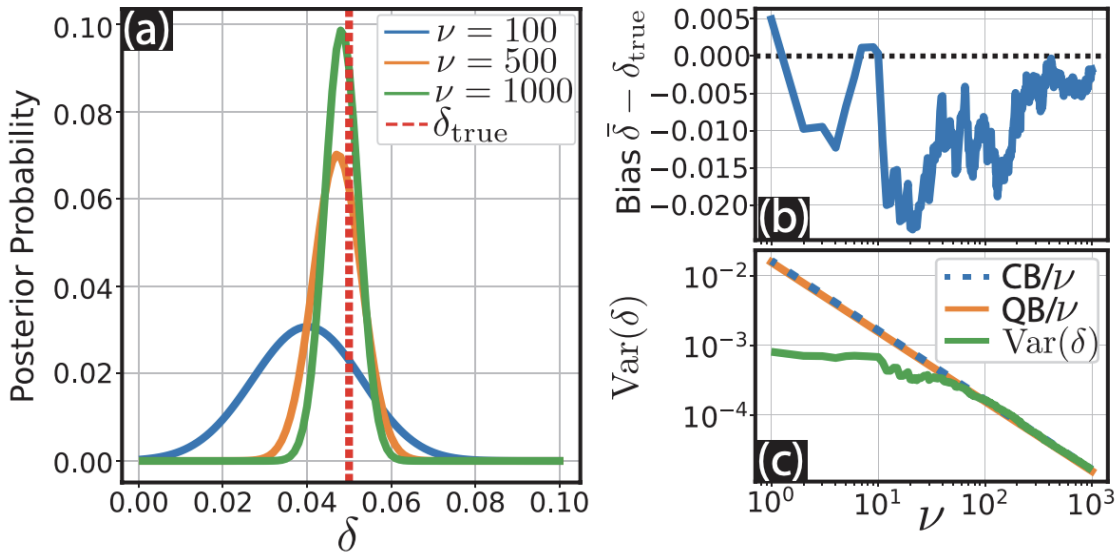


$$\nu\,\mathrm{Var}(\delta) \geq \boxed{\frac{1}{F(\delta)}} \geq \frac{1}{Q(\delta)},$$

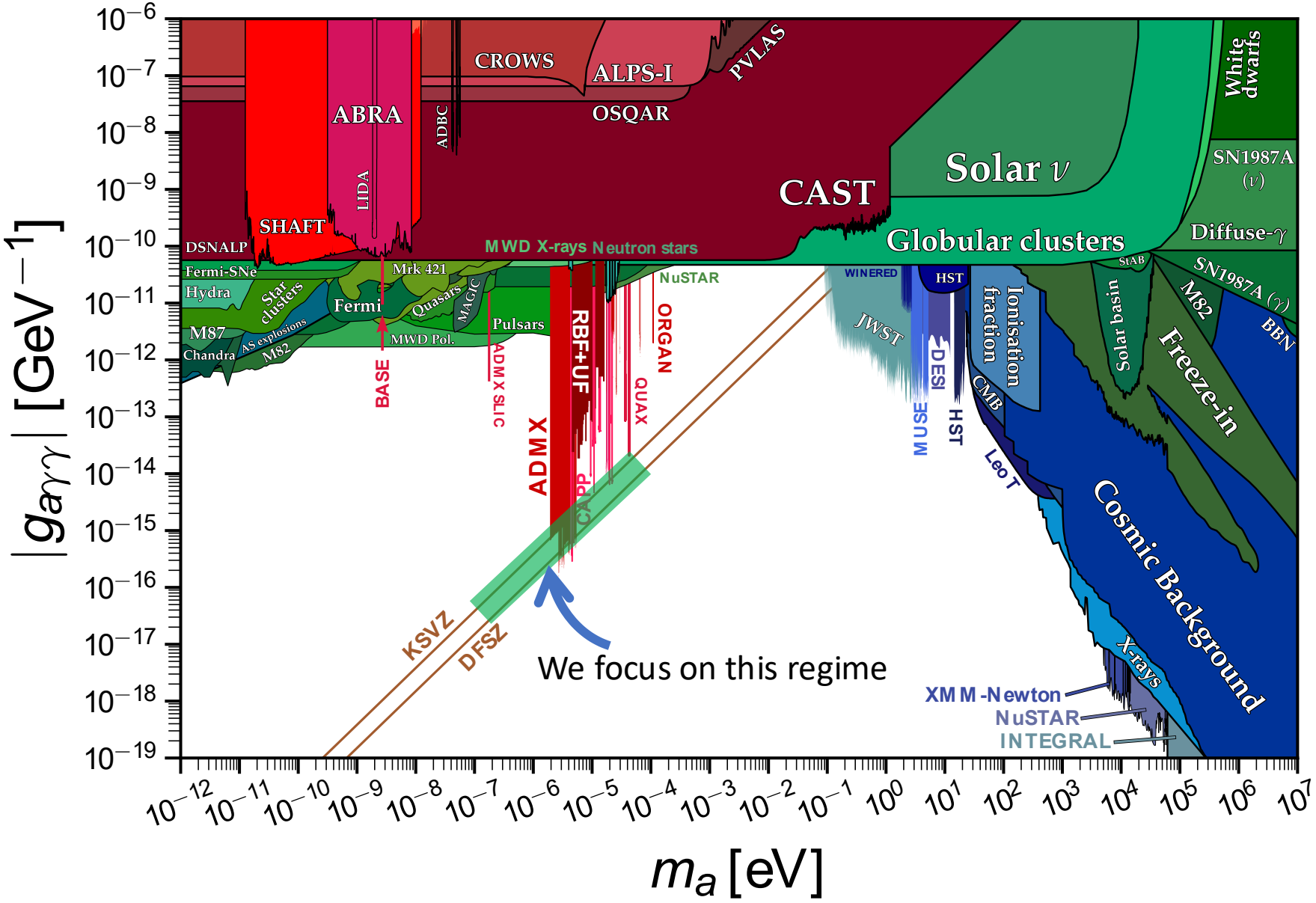
- Measurement ansatz fails to approximate optimal POVM.

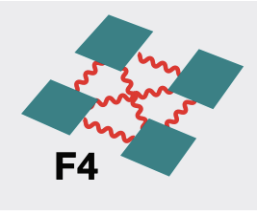
Bayesian estimation



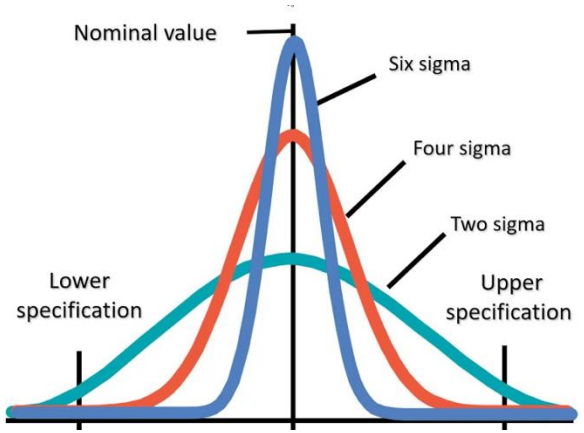


Future work





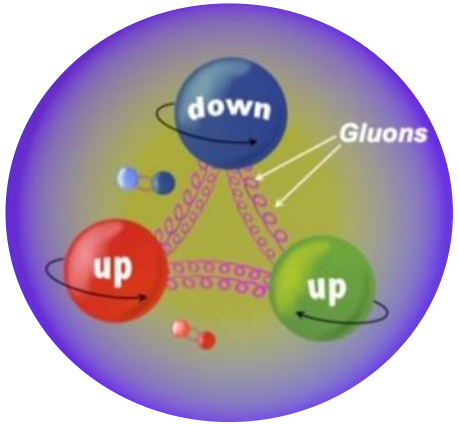
- Optimize at $m_a = 10^{-5}$
- Scan from 10^{-7} to 10^{-4}
- Apply Bayesian to evaluate mean and std to 6sigma

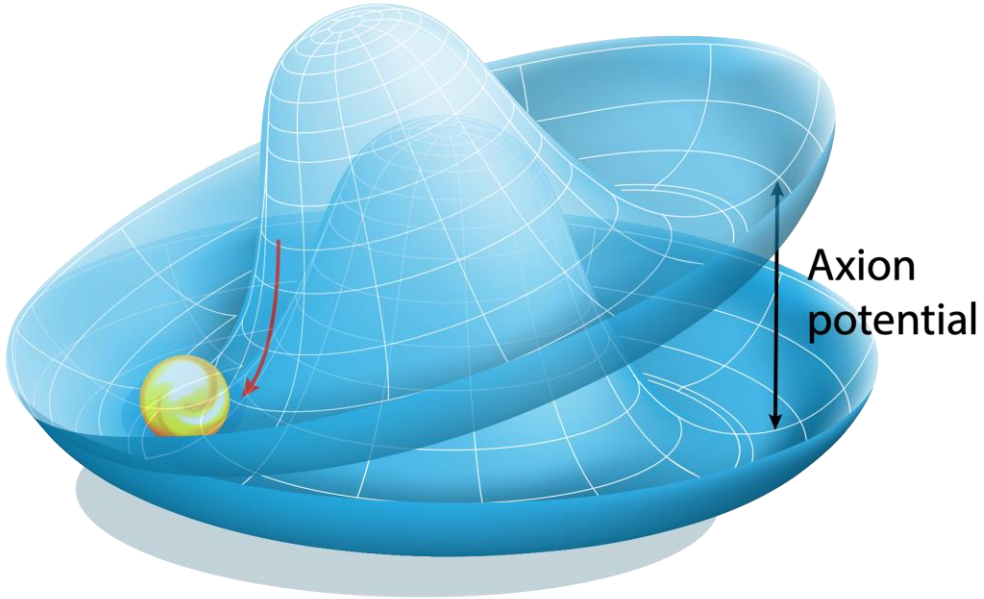


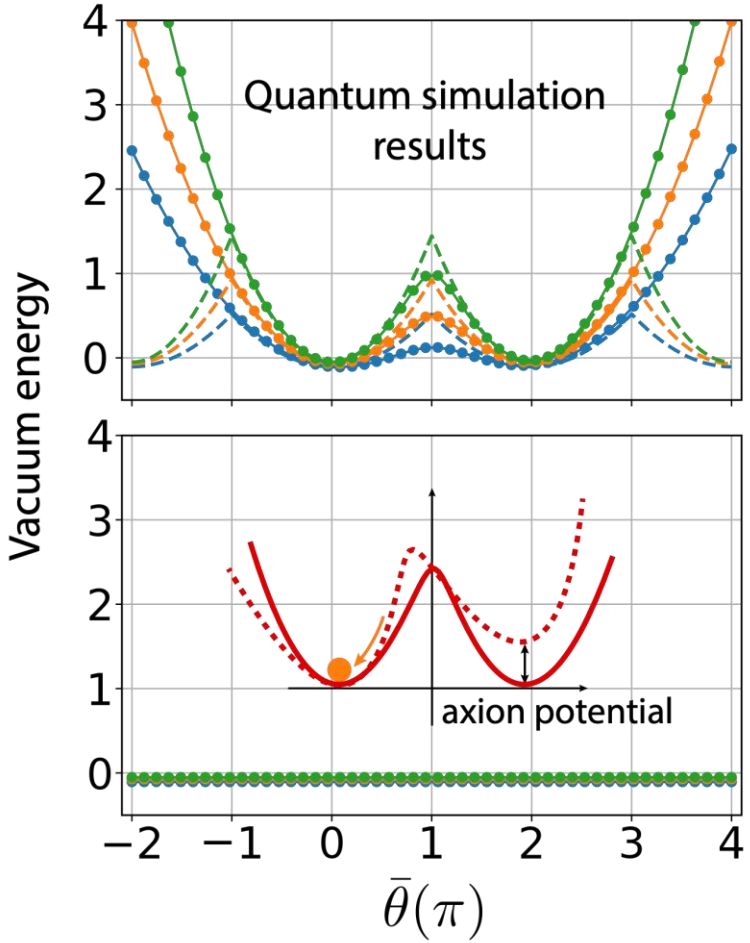
Future work: quantum simulation of strong CP problem

$$\mathcal{L}_{\mathrm{QCD}} = -\frac{1}{4} G^a_{\mu\nu} G^{a,\mu\nu} + \sum_f \bar{\psi}_f (i\gamma^\mu D_\mu - m_f)\psi_f$$

$$+ \bar{\theta} \frac{g^2}{32\pi^2} G^a_{\mu\nu} \tilde{G}^{a,\mu\nu}.$$







Le Bin Ho, Physical Review D 113, 074507 (2026)

Thank you very much for your attention!